# Decoder-free reproductions of magic-state simulations with real non-Clifford gates on a GPU

Generated by `repro/make_report.py`

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Every figure below is regenerated from scratch by `uv run python -m repro.run_all`: circuits come from the authors' generators, are sampled with the Tsim CUDA statevector executor (postselection on the device), and every point records its own GPU time, compile time, shot count, GPU model and Tsim build. Where the paper simulated a Clifford proxy ($S$ in place of $T$, or a $|Y\rangle$ / $|+i\rangle$ state), the same proxy is sampled again on the GPU and plotted next to the published numbers, as a check of the GPU kernel; the real non-Clifford circuit ($T$ gates, rotations) is sampled alongside. Logical error rates are per kept shot; bands and error bars span hypotheses within a factor 1000 of the maximum likelihood (the convention of `sinter`). Triangles mark upper bounds for points with zero observed errors.

Tsim builds: a2af975fbc.

| study | points | GPU sampling | shots |
|---|---|---|---|
| gidney_cultivation | 32 | 14.7 min | 2.4e12 |
| wan_few_clifford_terms | 16 | 4.5 min | 6.5e11 |
| soft_d5 | 1 | 12 s | 3.3e10 |
| clifft_inject_cultivate | 0 | 0.0 s | 0 |
| gauging_cultivation | 12 | 3.1 min | 5.4e11 |
| fold_transversal | 12 | 46.4 min | 3.7e12 |
| chan_flags | 28 | 19.9 min | 1.2e12 |
| magna | 36 | 25.3 min | 7.2e12 |
| google_experiment | 278 | 4.4 min | 1.5e12 |
| star_injection | 30 | 20.4 min | 7.6e12 |
| rp2_ungrown | 2 | 23 s | 7.9e10 |
| erasure_injection | 4 | 2.5 s | 2.9e9 |
| vaknin_full_vector | 4 | 48 s | 2.2e9 |

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**2 Wan, Zhong, Zapirain (2025), Simulating magic state cultivation with few Clifford terms** **8**

**3 Li et al. (2025), SOFT: a high-performance simulator for universal fault-tolerant quantum circuits** **10**

**4 Chase, Labib (2026), Clifft: fast exact simulation of near-Clifford quantum circuits** **11**

**5 Hetényi, Brown, Williamson (2026), Constant-depth magic state cultivation with Clifford measurements by gauging** **12**

**6 Sahay et al. (2025), Fold-transversal surface code cultivation** **13**

**7 Chan, Strikis, Sun, Cai (2026), Diagnosing and restoring the degraded fault distance of magic state cultivation 14**

**8 Wang, Gu (2026), Magna: simulating magic state factories as cheaply as Clifford circuits 16**

**9 Rosenfeld et al. (2025), Magic state cultivation on a superconducting quantum processor 18**

**10 Ismail et al. (2025), Transversal architecture for megaquop-scale quantum simulation with neutral atoms (transversal STAR) 20**

**11 Chen et al. (2025), Efficient magic state cultivation on $RP^2$ 21**

**12 Jacoby et al. (2025), Magic state injection with erasure qubits 22**

**13 Vaknin et al. (2025), Efficient magic state cultivation on the surface code 23**

# Reproduction checks against published numbers

Every published number (tables, released data) that a sampled point can be compared with. ±: one standard deviation (Poisson for error counts, binomial for acceptance; published values without an uncertainty, or exact values, have none). $z$: difference over the combined standard deviation.

| study | point | variant | quantity | published | Tsim GPU | ratio | $z$ |
|---|---|---|---|---|---|---|---|
| gidney_cultivation | bell d=3 p=0.0005 S (Stim) | clifford | LER | 4.91e-08 ± 2e-09 | 4.89e-08 ± 3e-10 | 0.996 | -0.1 |
| gidney_cultivation | bell d=3 p=0.0005 S (Stim) | clifford | acceptance | 0.846 ± 2e-06 | 0.846 ± 5e-07 | 1.000 | +1.2 |
| gidney_cultivation | bell d=3 p=0.001 S (Stim) | clifford | LER | 3.92e-07 ± 1e-08 | 3.99e-07 ± 3e-09 | 1.019 | +0.6 |
| gidney_cultivation | bell d=3 p=0.001 S (Stim) | clifford | acceptance | 0.716 ± 7e-06 | 0.716 ± 2e-06 | 1.000 | +2.1 |
| gidney_cultivation | bell d=3 p=0.002 S (Stim) | clifford | LER | 3.25e-06 ± 1e-07 | 3.32e-06 ± 2e-08 | 1.020 | +0.6 |
| gidney_cultivation | bell d=3 p=0.002 S (Stim) | clifford | acceptance | 0.513 ± 2e-05 | 0.513 ± 5e-06 | 1.000 | -1.5 |
| gidney_cultivation | teleport d=3 p=0.0005 S (Stim) | clifford | LER | 7.28e-08 ± 2e-09 | 7.12e-08 ± 5e-10 | 0.978 | -0.7 |
| gidney_cultivation | teleport d=3 p=0.0005 S (Stim) | clifford | acceptance | 0.804 ± 3e-06 | 0.804 ± 7e-07 | 1.000 | -1.7 |
| gidney_cultivation | teleport d=3 p=0.001 S (Stim) | clifford | LER | 5.79e-07 ± 2e-08 | 5.89e-07 ± 4e-09 | 1.018 | +0.6 |
| gidney_cultivation | teleport d=3 p=0.001 S (Stim) | clifford | acceptance | 0.646 ± 9e-06 | 0.646 ± 2e-06 | 1.000 | -0.9 |
| gidney_cultivation | teleport d=3 p=0.002 S (Stim) | clifford | LER | 4.81e-06 ± 2e-07 | 4.83e-06 ± 3e-08 | 1.005 | +0.2 |
| gidney_cultivation | teleport d=3 p=0.002 S (Stim) | clifford | acceptance | 0.418 ± 2e-05 | 0.418 ± 5e-06 | 1.000 | -2.3 |
| gidney_cultivation | unitary d=3 p=0.0005 S (Stim) | clifford | LER | 4e-08 ± 1e-09 | 4.11e-08 ± 3e-10 | 1.027 | +0.9 |
| gidney_cultivation | unitary d=3 p=0.0005 S (Stim) | clifford | acceptance | 0.829 ± 2e-06 | 0.829 ± 5e-07 | 1.000 | -1.6 |
| gidney_cultivation | unitary d=3 p=0.001 S (Stim) | clifford | LER | 3.42e-07 ± 1e-08 | 3.43e-07 ± 2e-09 | 1.001 | +0.0 |
| gidney_cultivation | unitary d=3 p=0.001 S (Stim) | clifford | acceptance | 0.687 ± 7e-06 | 0.687 ± 2e-06 | 1.000 | +1.0 |
| gidney_cultivation | unitary d=3 p=0.001 T (state vector) | t | LER | 9.73e-07 ± 5e-08 | 8.98e-07 ± 6e-09 | 0.923 | -1.4 |
| gidney_cultivation | unitary d=3 p=0.001 T (state vector) | t | acceptance | 0.687 ± 2e-05 | 0.687 ± 3e-06 | 1.000 | +0.7 |
| gidney_cultivation | unitary d=3 p=0.001 S (state vector) | clifford | LER | 3.89e-07 ± 3e-08 | 3.43e-07 ± 2e-09 | 0.880 | -1.4 |
| gidney_cultivation | unitary d=3 p=0.001 S (state vector) | clifford | acceptance | 0.687 ± 2e-05 | 0.687 ± 2e-06 | 1.000 | -1.4 |
| gidney_cultivation | unitary d=3 p=0.001 Z (state vector) | clifford | LER | 2.93e-07 ± 3e-08 | 2.88e-07 ± 2e-09 | 0.983 | -0.2 |
| gidney_cultivation | unitary d=3 p=0.001 Z (state vector) | clifford | acceptance | 0.687 ± 2e-05 | 0.687 ± 1e-06 | 1.000 | +2.2 |
| gidney_cultivation | unitary d=3 p=0.002 S (Stim) | clifford | LER | 2.85e-06 ± 9e-08 | 2.95e-06 ± 2e-08 | 1.037 | +1.1 |
| gidney_cultivation | unitary d=3 p=0.002 S (Stim) | clifford | acceptance | 0.472 ± 2e-05 | 0.472 ± 4e-06 | 1.000 | +0.3 |
| gidney_cultivation | unitary d=3 p=0.002 T (state vector) | t | LER | 5.86e-06 ± 1e-07 | 5.69e-06 ± 4e-08 | 0.972 | -1.1 |
| gidney_cultivation | unitary d=3 p=0.002 T (state vector) | t | acceptance | 0.472 ± 2e-05 | 0.472 ± 6e-06 | 1.000 | -0.3 |
| gidney_cultivation | unitary d=3 p=0.002 S (state vector) | clifford | LER | 2.87e-06 ± 1e-07 | 2.95e-06 ± 2e-08 | 1.028 | +0.7 |
| gidney_cultivation | unitary d=3 p=0.002 S (state vector) | clifford | acceptance | 0.472 ± 2e-05 | 0.472 ± 4e-06 | 1.000 | -1.1 |
| gidney_cultivation | unitary d=3 p=0.002 Z (state vector) | clifford | LER | 2.53e-06 ± 1e-07 | 2.46e-06 ± 2e-08 | 0.973 | -0.7 |
| gidney_cultivation | unitary d=3 p=0.002 Z (state vector) | clifford | acceptance | 0.472 ± 2e-05 | 0.472 ± 4e-06 | 1.000 | -1.1 |
| gidney_cultivation | unitary d=3 p=0.003 T (state vector) | t | LER | 1.81e-05 ± 6e-07 | 1.81e-05 ± 1e-07 | 1.000 | +0.0 |
| gidney_cultivation | unitary d=3 p=0.003 T (state vector) | t | acceptance | 0.325 ± 4e-05 | 0.325 ± 8e-06 | 1.000 | +1.2 |
| gidney_cultivation | unitary d=3 p=0.003 S (state vector) | clifford | LER | 1.03e-05 ± 4e-07 | 1.05e-05 ± 7e-08 | 1.027 | +0.7 |
| gidney_cultivation | unitary d=3 p=0.003 S (state vector) | clifford | acceptance | 0.325 ± 3e-05 | 0.325 ± 6e-06 | 1.000 | -1.4 |
| gidney_cultivation | unitary d=3 p=0.003 Z (state vector) | clifford | LER | 9.28e-06 ± 4e-07 | 9e-06 ± 6e-08 | 0.970 | -0.7 |
| gidney_cultivation | unitary d=3 p=0.003 Z (state vector) | clifford | acceptance | 0.325 ± 3e-05 | 0.325 ± 6e-06 | 1.000 | +1.1 |
| gidney_cultivation | unitary d=3 p=0.005 T (state vector) | t | LER | 8.75e-05 ± 3e-06 | 8.48e-05 ± 4e-07 | 0.969 | -1.0 |
| gidney_cultivation | unitary d=3 p=0.005 T (state vector) | t | acceptance | 0.155 ± 4e-05 | 0.155 ± 6e-06 | 0.999 | -2.6 |
| gidney_cultivation | unitary d=3 p=0.005 S (state vector) | clifford | LER | 5.81e-05 ± 2e-06 | 5.69e-05 ± 3e-07 | 0.980 | -0.6 |
| gidney_cultivation | unitary d=3 p=0.005 S (state vector) | clifford | acceptance | 0.155 ± 3e-05 | 0.155 ± 6e-06 | 1.000 | +0.6 |
| gidney_cultivation | unitary d=3 p=0.005 Z (state vector) | clifford | LER | 4.92e-05 ± 2e-06 | 4.84e-05 ± 3e-07 | 0.983 | -0.5 |
| gidney_cultivation | unitary d=3 p=0.005 Z (state vector) | clifford | acceptance | 0.155 ± 3e-05 | 0.155 ± 6e-06 | 1.000 | -0.3 |
| gidney_cultivation | unitary d=3 p=0.007 T (state vector) | t | LER | 0.00026 ± 8e-06 | 0.000252 ± 2e-06 | 0.970 | -0.9 |
| gidney_cultivation | unitary d=3 p=0.007 T (state vector) | t | acceptance | 0.0741 ± 4e-05 | 0.0741 ± 7e-06 | 1.001 | +1.3 |
| gidney_cultivation | unitary d=3 p=0.007 S (state vector) | clifford | LER | 0.00019 ± 6e-06 | 0.000184 ± 7e-07 | 0.966 | -1.1 |
| gidney_cultivation | unitary d=3 p=0.007 S (state vector) | clifford | acceptance | 0.0741 ± 3e-05 | 0.0742 ± 3e-06 | 1.000 | +0.6 |
| gidney_cultivation | unitary d=3 p=0.007 Z (state vector) | clifford | LER | 0.000154 ± 5e-06 | 0.000154 ± 6e-07 | 1.003 | +0.1 |
| gidney_cultivation | unitary d=3 p=0.007 Z (state vector) | clifford | acceptance | 0.0743 ± 3e-05 | 0.0743 ± 3e-06 | 1.000 | +0.9 |
| gidney_cultivation | unitary d=3 p=0.01 T (state vector) | t | LER | 0.00091 ± 3e-05 | 0.000866 ± 5e-06 | 0.951 | -1.5 |
| gidney_cultivation | unitary d=3 p=0.01 T (state vector) | t | acceptance | 0.0249 ± 2e-05 | 0.0249 ± 4e-06 | 0.998 | -1.7 |
| gidney_cultivation | unitary d=3 p=0.01 S (state vector) | clifford | LER | 0.000667 ± 2e-05 | 0.000685 ± 4e-06 | 1.028 | +0.9 |
| gidney_cultivation | unitary d=3 p=0.01 S (state vector) | clifford | acceptance | 0.0249 ± 2e-05 | 0.0249 ± 4e-06 | 1.001 | +1.4 |
| gidney_cultivation | unitary d=3 p=0.01 Z (state vector) | clifford | LER | 0.000573 ± 2e-05 | 0.000585 ± 4e-06 | 1.021 | +0.7 |
| gidney_cultivation | unitary d=3 p=0.01 Z (state vector) | clifford | acceptance | 0.025 ± 2e-05 | 0.025 ± 4e-06 | 1.000 | +0.4 |

| study | point | variant | quantity | published | Tsim GPU | ratio | $z$ |
|---|---|---|---|---|---|---|---|
| wan_few_clifford_terms | d=3 p=0.0005 T | t | LER | 1.8e-07 ± 7e-09 | 1.83e-07 ± 1e-09 | 1.022 | +0.5 |
| wan_few_clifford_terms | d=3 p=0.0005 T | t | acceptance | 0.807 ± 6e-06 | 0.804 ± 1e-06 | 0.997 | -375.0 |
| wan_few_clifford_terms | d=3 p=0.001 T | t | LER | 1e-06 ± 5e-08 | 1.06e-06 ± 7e-09 | 1.055 | +1.0 |
| wan_few_clifford_terms | d=3 p=0.001 T | t | acceptance | 0.651 ± 2e-05 | 0.647 ± 3e-06 | 0.994 | -179.4 |
| wan_few_clifford_terms | d=3 p=0.0015 T | t | LER | 2.95e-06 ± 1e-07 | 3.09e-06 ± 2e-08 | 1.049 | +1.4 |
| wan_few_clifford_terms | d=3 p=0.0015 T | t | acceptance | 0.525 ± 2e-05 | 0.521 ± 4e-06 | 0.991 | -204.1 |
| wan_few_clifford_terms | d=3 p=0.002 T | t | LER | 6.76e-06 ± 2e-07 | 6.94e-06 ± 5e-08 | 1.027 | +1.0 |
| wan_few_clifford_terms | d=3 p=0.002 T | t | acceptance | 0.424 ± 2e-05 | 0.419 ± 6e-06 | 0.989 | -217.6 |
| wan_few_clifford_terms | d=3 p=0.0025 T | t | LER | 1.29e-05 ± 3e-07 | 1.28e-05 ± 9e-08 | 0.992 | -0.4 |
| wan_few_clifford_terms | d=3 p=0.0025 T | t | acceptance | 0.343 ± 2e-05 | 0.338 ± 7e-06 | 0.986 | -224.8 |
| wan_few_clifford_terms | d=3 p=0.003 T | t | LER | 2.15e-05 ± 4e-07 | 2.17e-05 ± 2e-07 | 1.008 | +0.4 |
| wan_few_clifford_terms | d=3 p=0.003 T | t | acceptance | 0.277 ± 2e-05 | 0.272 ± 8e-06 | 0.983 | -227.5 |
| wan_few_clifford_terms | d=3 p=0.004 T | t | LER | 4.93e-05 ± 7e-07 | 5.08e-05 ± 3e-07 | 1.030 | +1.9 |
| wan_few_clifford_terms | d=3 p=0.004 T | t | acceptance | 0.181 ± 2e-05 | 0.177 ± 7e-06 | 0.978 | -222.8 |
| wan_few_clifford_terms | d=3 p=0.005 T | t | LER | 9.77e-05 ± 1e-06 | 9.9e-05 ± 5e-07 | 1.012 | +0.9 |
| wan_few_clifford_terms | d=3 p=0.005 T | t | acceptance | 0.118 ± 1e-05 | 0.115 ± 6e-06 | 0.972 | -219.7 |
| gauging_cultivation | d=3 p=0.001 \|+i> proxy | clifford | LER | 8.8e-07 ± 8e-08 | 8.77e-07 ± 6e-09 | 0.996 | -0.0 |
| gauging_cultivation | d=3 p=0.001 \|+i> proxy | clifford | acceptance | 0.46 | 0.461 ± 2e-06 | 1.002 | +334.7 |
| gauging_cultivation | d=3 p=0.0005 \|+i> proxy | clifford | LER | 1.2e-07 ± 2e-08 | 1.07e-07 ± 8e-10 | 0.891 | -0.7 |
| gauging_cultivation | d=3 p=0.0005 \|+i> proxy | clifford | acceptance | 0.65 | 0.679 ± 9e-07 | 1.044 | +32201.3 |
| rp2_ungrown | MSC3 S proxy | clifford | LER | 8.3e-07 | 8.25e-07 ± 6e-09 | 0.995 | -0.8 |
| rp2_ungrown | MSC3 S proxy | clifford | acceptance | 0.514 | 0.514 ± 2e-06 | 1.000 | +14.7 |

## GPU kernel vs. Stim on the same Clifford circuits

Every Clifford-proxy point re-sampled with Stim on the CPU (`repro/stim_check.py`), same circuit file and postselection: an independent test of the Tsim GPU kernel.

| study | point | quantity | Stim (CPU) | Tsim GPU | Stim shots | $z$ |
|---|---|---|---|---|---|---|
| gidney_cultivation | bell_d3_p0.0005_s | acceptance | 0.846 ± 1e-05 | 0.846 ± 5e-07 | 1e9 | -0.5 |
| gidney_cultivation | bell_d3_p0.0005_s | LER | 4.96e-08 ± 8e-09 | 4.89e-08 ± 3e-10 | 1e9 | -0.1 |
| gidney_cultivation | bell_d3_p0.001_s | acceptance | 0.716 ± 1e-05 | 0.716 ± 2e-06 | 1e9 | -0.4 |
| gidney_cultivation | bell_d3_p0.001_s | LER | 3.55e-07 ± 2e-08 | 3.99e-07 ± 3e-09 | 1e9 | +2.0 |
| gidney_cultivation | bell_d3_p0.002_s | acceptance | 0.513 ± 2e-05 | 0.513 ± 5e-06 | 1e9 | -0.2 |
| gidney_cultivation | bell_d3_p0.002_s | LER | 3.29e-06 ± 8e-08 | 3.32e-06 ± 2e-08 | 1e9 | +0.4 |
| gidney_cultivation | teleport_d3_p0.0005_s | acceptance | 0.804 ± 1e-05 | 0.804 ± 7e-07 | 1e9 | +0.2 |
| gidney_cultivation | teleport_d3_p0.0005_s | LER | 7.59e-08 ± 1e-08 | 7.12e-08 ± 5e-10 | 1e9 | -0.5 |
| gidney_cultivation | teleport_d3_p0.001_s | acceptance | 0.646 ± 2e-05 | 0.646 ± 2e-06 | 1e9 | -1.7 |
| gidney_cultivation | teleport_d3_p0.001_s | LER | 5.42e-07 ± 3e-08 | 5.89e-07 ± 4e-09 | 1e9 | +1.6 |
| gidney_cultivation | teleport_d3_p0.002_s | acceptance | 0.418 ± 2e-05 | 0.418 ± 5e-06 | 1e9 | -0.6 |
| gidney_cultivation | teleport_d3_p0.002_s | LER | 4.86e-06 ± 1e-07 | 4.83e-06 ± 3e-08 | 1e9 | -0.3 |
| gidney_cultivation | unitary_d3_p0.0005_s | acceptance | 0.829 ± 1e-05 | 0.829 ± 5e-07 | 1e9 | -2.0 |
| gidney_cultivation | unitary_d3_p0.0005_s | LER | 3.14e-08 ± 6e-09 | 4.11e-08 ± 3e-10 | 1e9 | +1.6 |
| gidney_cultivation | unitary_d3_p0.001_s | acceptance | 0.687 ± 1e-05 | 0.687 ± 2e-06 | 1e9 | -0.3 |
| gidney_cultivation | unitary_d3_p0.001_s | LER | 3.54e-07 ± 2e-08 | 3.43e-07 ± 2e-09 | 1e9 | -0.5 |
| gidney_cultivation | unitary_d3_p0.001_z | acceptance | 0.687 ± 1e-05 | 0.687 ± 1e-06 | 1e9 | +1.0 |
| gidney_cultivation | unitary_d3_p0.001_z | LER | 2.87e-07 ± 2e-08 | 2.88e-07 ± 2e-09 | 1e9 | +0.1 |
| gidney_cultivation | unitary_d3_p0.002_s | acceptance | 0.472 ± 2e-05 | 0.472 ± 4e-06 | 1e9 | +1.1 |
| gidney_cultivation | unitary_d3_p0.002_s | LER | 2.98e-06 ± 8e-08 | 2.95e-06 ± 2e-08 | 1e9 | -0.3 |
| gidney_cultivation | unitary_d3_p0.002_z | acceptance | 0.472 ± 2e-05 | 0.472 ± 4e-06 | 1e9 | -0.5 |
| gidney_cultivation | unitary_d3_p0.002_z | LER | 2.53e-06 ± 7e-08 | 2.46e-06 ± 2e-08 | 1e9 | -0.8 |
| gidney_cultivation | unitary_d3_p0.003_s | acceptance | 0.325 ± 1e-05 | 0.325 ± 6e-06 | 1e9 | +0.1 |
| gidney_cultivation | unitary_d3_p0.003_s | LER | 1.04e-05 ± 2e-07 | 1.05e-05 ± 7e-08 | 1e9 | +0.5 |
| gidney_cultivation | unitary_d3_p0.003_z | acceptance | 0.325 ± 1e-05 | 0.325 ± 6e-06 | 1e9 | +0.9 |
| gidney_cultivation | unitary_d3_p0.003_z | LER | 8.92e-06 ± 2e-07 | 9e-06 ± 6e-08 | 1e9 | +0.5 |
| gidney_cultivation | unitary_d3_p0.005_s | acceptance | 0.155 ± 1e-05 | 0.155 ± 6e-06 | 1e9 | +0.3 |
| gidney_cultivation | unitary_d3_p0.005_s | LER | 5.65e-05 ± 6e-07 | 5.69e-05 ± 3e-07 | 1e9 | +0.6 |
| gidney_cultivation | unitary_d3_p0.005_z | acceptance | 0.155 ± 1e-05 | 0.155 ± 6e-06 | 1e9 | -0.0 |

| study | point | quantity | Stim (CPU) | Tsim GPU | Stim shots | $z$ |
|---|---|---|---|---|---|---|
| gidney_cultivation | unitary_d3_p0.005_z | LER | 4.84e-05 ± 6e-07 | 4.84e-05 ± 3e-07 | 1e9 | -0.0 |
| gidney_cultivation | unitary_d3_p0.007_s | acceptance | 0.0742 ± 8e-06 | 0.0742 ± 3e-06 | 1e9 | +1.0 |
| gidney_cultivation | unitary_d3_p0.007_s | LER | 0.000183 ± 2e-06 | 0.000184 ± 7e-07 | 1e9 | +0.8 |
| gidney_cultivation | unitary_d3_p0.007_z | acceptance | 0.0743 ± 8e-06 | 0.0743 ± 3e-06 | 1e9 | +1.7 |
| gidney_cultivation | unitary_d3_p0.007_z | LER | 0.000156 ± 1e-06 | 0.000154 ± 6e-07 | 1e9 | -1.0 |
| gidney_cultivation | unitary_d3_p0.01_s | acceptance | 0.0249 ± 5e-06 | 0.0249 ± 4e-06 | 1e9 | +0.5 |
| gidney_cultivation | unitary_d3_p0.01_s | LER | 0.000692 ± 5e-06 | 0.000685 ± 4e-06 | 1e9 | -1.0 |
| gidney_cultivation | unitary_d3_p0.01_z | acceptance | 0.025 ± 5e-06 | 0.025 ± 4e-06 | 1e9 | -0.2 |
| gidney_cultivation | unitary_d3_p0.01_z | LER | 0.000579 ± 5e-06 | 0.000585 ± 4e-06 | 1e9 | +0.8 |
| wan_few_clifford_terms | d3_p0.0005_s | acceptance | 0.804 ± 1e-05 | 0.804 ± 7e-07 | 1e9 | +0.0 |
| wan_few_clifford_terms | d3_p0.0005_s | LER | 5.97e-08 ± 9e-09 | 7.18e-08 ± 5e-10 | 1e9 | +1.4 |
| wan_few_clifford_terms | d3_p0.0015_s | acceptance | 0.519 ± 2e-05 | 0.519 ± 4e-06 | 1e9 | +1.3 |
| wan_few_clifford_terms | d3_p0.0015_s | LER | 2.01e-06 ± 6e-08 | 2.02e-06 ± 1e-08 | 1e9 | +0.1 |
| wan_few_clifford_terms | d3_p0.001_s | acceptance | 0.646 ± 2e-05 | 0.646 ± 2e-06 | 1e9 | -0.7 |
| wan_few_clifford_terms | d3_p0.001_s | LER | 5.65e-07 ± 3e-08 | 5.81e-07 ± 4e-09 | 1e9 | +0.5 |
| wan_few_clifford_terms | d3_p0.0025_s | acceptance | 0.336 ± 1e-05 | 0.336 ± 6e-06 | 1e9 | +0.6 |
| wan_few_clifford_terms | d3_p0.0025_s | LER | 9.64e-06 ± 2e-07 | 9.67e-06 ± 7e-08 | 1e9 | +0.1 |
| wan_few_clifford_terms | d3_p0.002_s | acceptance | 0.418 ± 2e-05 | 0.418 ± 5e-06 | 1e9 | +1.4 |
| wan_few_clifford_terms | d3_p0.002_s | LER | 4.87e-06 ± 1e-07 | 4.85e-06 ± 3e-08 | 1e9 | -0.2 |
| wan_few_clifford_terms | d3_p0.003_s | acceptance | 0.27 ± 1e-05 | 0.27 ± 6e-06 | 1e9 | +0.3 |
| wan_few_clifford_terms | d3_p0.003_s | LER | 1.68e-05 ± 2e-07 | 1.69e-05 ± 1e-07 | 1e9 | +0.4 |
| wan_few_clifford_terms | d3_p0.004_s | acceptance | 0.175 ± 1e-05 | 0.175 ± 6e-06 | 1e9 | -0.6 |
| wan_few_clifford_terms | d3_p0.004_s | LER | 4.2e-05 ± 5e-07 | 4.21e-05 ± 2e-07 | 1e9 | +0.3 |
| wan_few_clifford_terms | d3_p0.005_s | acceptance | 0.114 ± 1e-05 | 0.114 ± 6e-06 | 1e9 | -0.6 |
| wan_few_clifford_terms | d3_p0.005_s | LER | 8.56e-05 ± 9e-07 | 8.47e-05 ± 5e-07 | 1e9 | -0.9 |
| gauging_cultivation | d3_p0.0005_s | acceptance | 0.679 ± 1e-05 | 0.679 ± 9e-07 | 1e9 | -0.1 |
| gauging_cultivation | d3_p0.0005_s | LER | 9.73e-08 ± 1e-08 | 1.07e-07 ± 8e-10 | 1e9 | +0.8 |
| gauging_cultivation | d3_p0.001_s | acceptance | 0.461 ± 2e-05 | 0.461 ± 2e-06 | 1e9 | -0.9 |
| gauging_cultivation | d3_p0.001_s | LER | 9.12e-07 ± 4e-08 | 8.77e-07 ± 6e-09 | 1e9 | -0.8 |
| gauging_cultivation | d3_p0.002_s | acceptance | 0.213 ± 1e-05 | 0.213 ± 4e-06 | 1e9 | +0.2 |
| gauging_cultivation | d3_p0.002_s | LER | 7.7e-06 ± 2e-07 | 7.58e-06 ± 5e-08 | 1e9 | -0.6 |
| gauging_cultivation | d3_p0.003_s | acceptance | 0.0983 ± 9e-06 | 0.0983 ± 3e-06 | 1e9 | +1.1 |
| gauging_cultivation | d3_p0.003_s | LER | 2.65e-05 ± 5e-07 | 2.76e-05 ± 2e-07 | 1e9 | +1.9 |
| gauging_cultivation | d3_p0.004_s | acceptance | 0.0456 ± 7e-06 | 0.0456 ± 2e-06 | 1e9 | +0.7 |
| gauging_cultivation | d3_p0.004_s | LER | 6.92e-05 ± 1e-06 | 7.04e-05 ± 4e-07 | 1e9 | +1.0 |
| gauging_cultivation | d3_p0.005_s | acceptance | 0.0211 ± 5e-06 | 0.0212 ± 1e-06 | 1e9 | +1.1 |
| gauging_cultivation | d3_p0.005_s | LER | 0.000147 ± 3e-06 | 0.000148 ± 7e-07 | 1e9 | +0.3 |
| fold_transversal | fd3_p0.0005_y | acceptance | 0.897 ± 1e-05 | 0.897 ± 3e-07 | 1e9 | -0.4 |
| fold_transversal | fd3_p0.0005_y | LER | 7.81e-09 ± 3e-09 | 6.34e-09 ± 8e-11 | 1e9 | -0.5 |
| fold_transversal | fd3_p0.0007_y | acceptance | 0.858 ± 1e-05 | 0.858 ± 3e-07 | 1e9 | +0.1 |
| fold_transversal | fd3_p0.0007_y | LER | 2.45e-08 ± 5e-09 | 1.74e-08 ± 1e-10 | 1e9 | -1.3 |

# 1 Gidney, Shutty, Jones (2024), Magic state cultivation: growing T states as cheap as CNOT gates

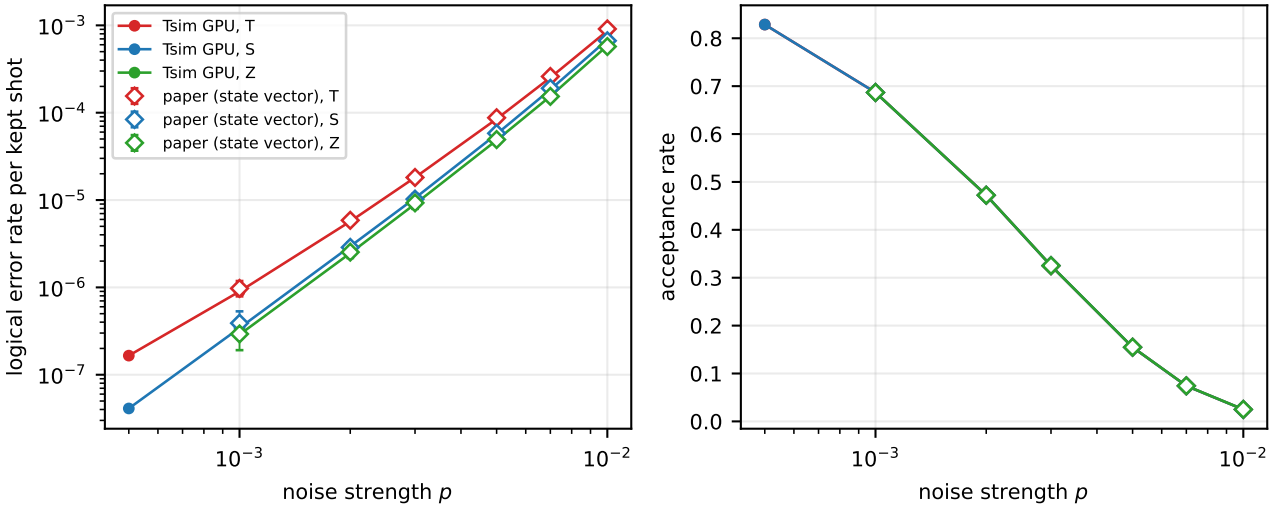
**arXiv:2409.17595** Study module `repro/studies/gidney_cultivation.py`

Two figures, both decoder-free (all detectors postselected, noiseless final logical check):

- Fig. 13 (`t-vs-s`): unitary injection + d=3 cultivation with each T gate replaced by T, S or Z, vs. uniform depolarizing noise p. The paper used a state-vector simulator ("vec_intercept").
- Fig. 21 (`inject-only-sampled`): bell / teleport ("degenerate") / unitary injection + d=3 cultivation, and unitary + d=5 cultivation, at p = 5e-4, 1e-3, 2e-3. The paper sampled the S proxy with Stim ("perfectionist" decoder = discard on any detection event).

Circuits: the authors' generator (`cultiv.make_inject_and_cultivate_circuit`, commit 871e68f) with their `gen.NoiseModel.uniform_depolarizing(p)`; T circuits by the Clifft paper's converter (`convert_circuit`: S->T, d=5 sign flips and feedforward, final Y_L wrapped in T...T_DAG). Z variant: S/S_DAG -> Z and the final logical check measures X_L instead of Y_L.

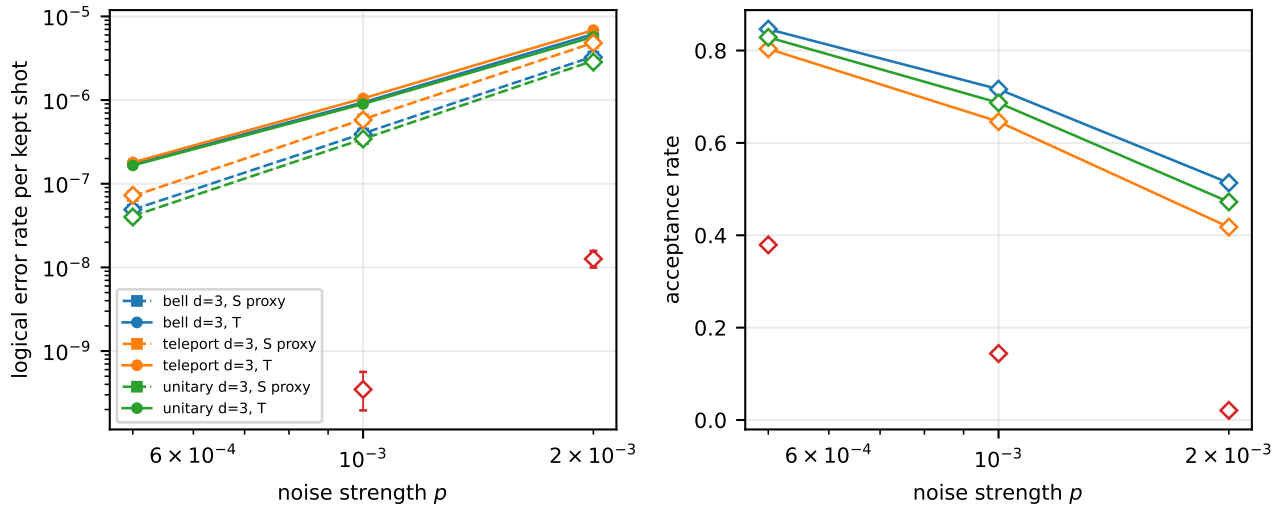
Published numbers overlaid: Zenodo 13777072 `stats.csv`.




**Gidney et al. Fig. 13: cultivating $T$, $S$, $Z$ states ($d = 3$, no escape)**

*Compute: 1.6 min GPU sampling + 5.3 s compile, 3.1e11 shots, 18 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)*

Unitary injection followed by $d = 3$ cultivation; every detector postselected; noiseless final logical check (for $T$: $T^\dagger$-rotated $Y_L$). Open markers: the paper's state-vector simulation (Zenodo stats.csv). Filled markers: Tsim GPU. $S$ and $Z$ are Clifford proxies of the same circuit.




**Gidney et al. Fig. 21: injection + cultivation, sampled**
*Compute: 14.0 min GPU sampling + 6.0 s compile, 2.2e12 shots, 18 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)*
Three injection styles with $d = 3$ cultivation and unitary injection with $d = 5$ cultivation, all detectors postselected. Dashed/squares: the $S$ proxy (the paper's Clifford simulation); solid/circles: real $T$ gates. Open diamonds: the paper's Stim data (Zenodo stats.csv, up to $10^{12}$ shots per point). The right panel shows discard rates of the $S$ proxy.

**Gidney et al.: $d = 5$ injection + cultivation (no escape)**
Each cell: logical error rate of kept shots, errors / kept shots, discard rate, and (Tsim) GPU sampling time.
*Compute: not yet sampled*

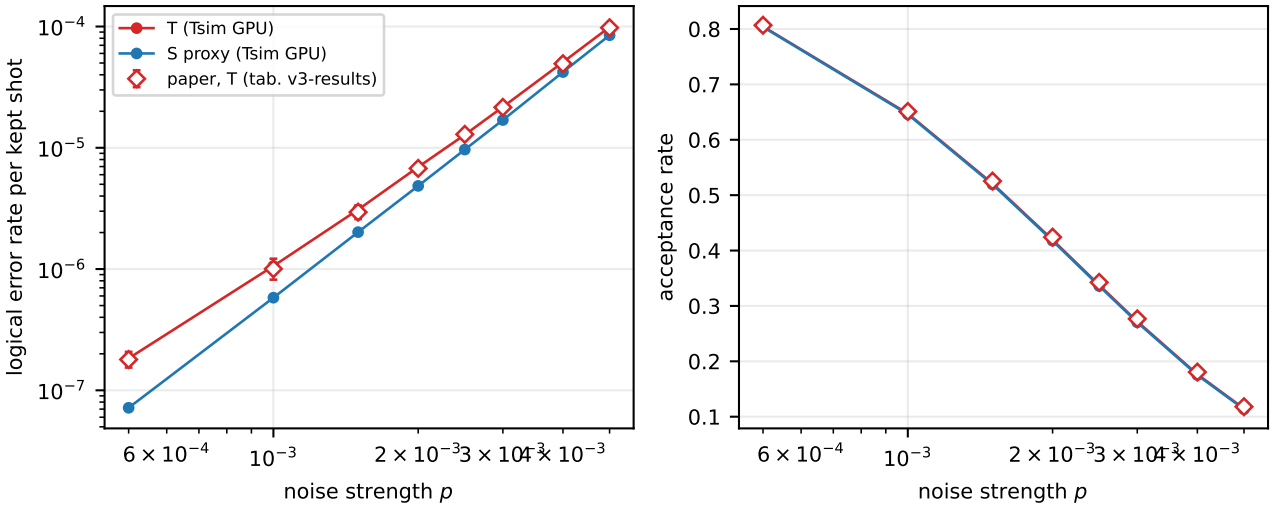
| $p$ | paper ($S$ proxy, Stim) | Tsim GPU, $S$ proxy | Tsim GPU, $T$ |
|---|---|---|---|
| 0.0005 | 0 (0/3.8e11, disc. 0.6208) | – | – |
| 0.001 | 3.47e-10 (50/1.4e11, disc. 0.8560) | – | – |
| 0.002 | 1.26e-08 (263/2.1e10, disc. 0.9792) | – | – |

# 2 Wan, Zhong, Zapirain (2025), Simulating magic state cultivation with few Clifford terms

**arXiv:2509.08658** Study module `repro/studies/wan_few_clifford_terms.py`

The paper samples the native-$T$ $d = 3$ cultivation circuit (injection + double check, from the companion repo's `d_3_circuit_definitions.py`) followed by a noiseless $T$-projection readout, postselecting on every detector. Its Table (`tab:v3-results`) lists the exact non-Clifford results at $2^{29}$–$2^{32}$ shots per noise strength; its red curves are the $S$-gate proxy sampled with Stim.

Circuits: built as in the repo's `run.py` and our `make_circuits.py`: $T \to S$, the vendored `gen.NoiseModel.uniform_depola` $S \to T$; the projection stays noiseless. The Clifford proxy maps every $T/T^\dagger$ of the whole circuit (including the projection) to $S/S^\dagger$.




**Wan et al. Figs. 15–17, 19: $d = 3$ $T$ cultivation, LER and post-selection rate**

*Compute: 4.5 min GPU sampling + 5.3 s compile, 6.5e11 shots, 16 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)*

Injection + double check with a noiseless $T$-projection readout; every detector postselected. Open diamonds: the paper's exact non-Clifford results (Table v3-results, $2^{29}$–$2^{32}$ shots, Apple M4 Pro CPU). Red: real $T$ gates; blue: the $S$ proxy (the paper's Stim reference). The LERs agree with the table; the published post-selection rates are 0.3–0.6% higher than ours, also for the $S$ proxy sampled with Stim ($10^8$ shots: 0.80367 vs. Tsim GPU 0.80365 at $p = 5 \times 10^{-4}$), so the circuit the repo generates is slightly different from the one behind the table. The $T$ circuit keeps about 0.1% more shots than its $S$ proxy.

**Wan et al. Table v3-results vs. Tsim GPU**

*Compute: 1.3 min GPU sampling + 3.0 s compile, 2e11 shots, 8 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)*

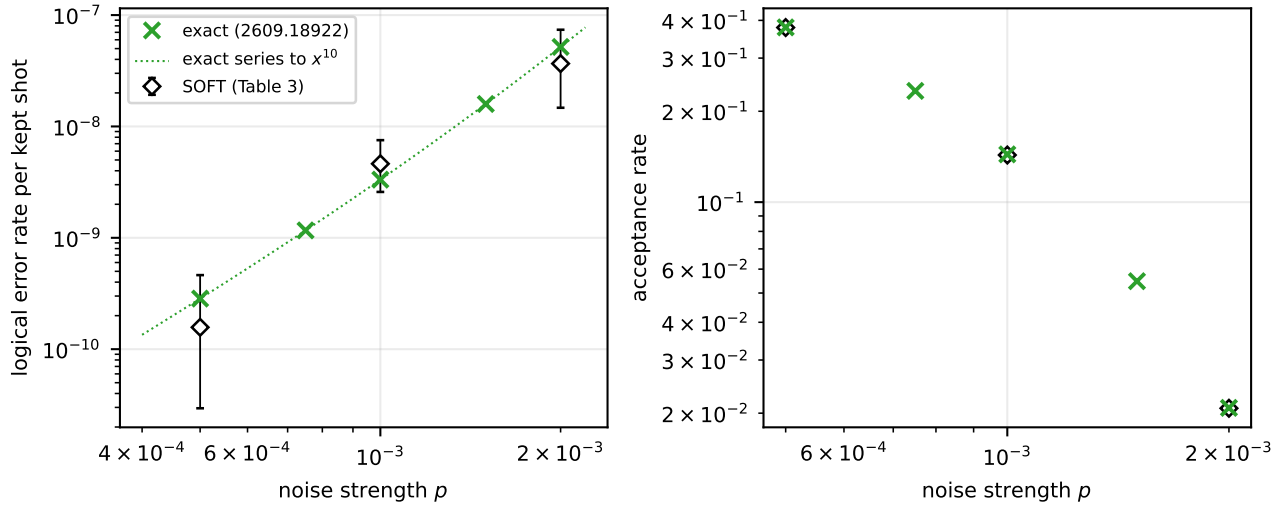
| | paper ($T$, CPU) | | | | Tsim GPU ($T$) | | | | |
|---|---|---|---|---|---|---|---|---|---|
| $p$ | LER | PSR | errors | shots | LER | PSR | errors | shots | time |
| 0.0005 | 1.8e-07 | 0.807 | 622 | 4.3e9 | 1.83e-07 | 0.8045 | 20029 | 1.4e11 | 60 s |
| 0.001 | 1e-06 | 0.651 | 351 | 5.4e8 | 1.06e-06 | 0.6473 | 20034 | 2.9e10 | 11 s |
| 0.0015 | 2.9e-06 | 0.525 | 831 | 5.4e8 | 3.09e-06 | 0.5210 | 20479 | 1.3e10 | 4.1 s |
| 0.002 | 6.8e-06 | 0.424 | 1540 | 5.4e8 | 6.94e-06 | 0.4194 | 20425 | 7e9 | 2.1 s |
| 0.0025 | 1.3e-05 | 0.343 | 2373 | 5.4e8 | 1.28e-05 | 0.3377 | 20794 | 4.8e9 | 1.3 s |
| 0.003 | 2.2e-05 | 0.277 | 3201 | 5.4e8 | 2.17e-05 | 0.2720 | 20768 | 3.5e9 | 1.0 s |
| 0.004 | 4.9e-05 | 0.181 | 4785 | 5.4e8 | 5.08e-05 | 0.1766 | 23758 | 2.6e9 | 0.8 s |
| 0.005 | 9.8e-05 | 0.118 | 6193 | 5.4e8 | 9.9e-05 | 0.1147 | 37503 | 3.3e9 | 0.8 s |

# 3 Li et al. (2025), SOFT: a high-performance simulator for universal fault-tolerant quantum circuits

**arXiv:2512.23037** Study module `repro/studies/soft_d5.py`

SOFT sampled the authors' own native-$T$ circuits (unitary injection, $d = 3$ and $d = 5$ double checks, code growth, noiseless readout; every detector postselected) on 16 H800 GPUs for 20 days: 28.9B, 74B and 134.4B shots at $p$ = 2e-3, 1e-3, 5e-4. SOFT ran no Clifford proxy, so only the non-Clifford version is sampled here (the $S$ proxy of the same protocol is in the Gidney study).

Circuits: SOFT's released files (`softv1_magic_state_cultivation`), used verbatim. They are also the reference circuits of the exact-LER paper (Wan, Zapirain, arXiv:2609.18922), whose exact values $A(p)$ and $P_L(p)$ and truncated series are overlaid.




**SOFT Fig. 2 / Table 3: $d = 5$ injection + cultivation with real $T$ gates**
*Compute: not yet sampled*
SOFT's own circuits (verbatim), all detectors postselected, noiseless final check. Black diamonds: SOFT's published values (16 H800 GPUs, 20 days, $2.4 \times 10^{11}$ shots). Green crosses and dotted line: exact values and series of Wan and Zapirain (arXiv:2609.18922) for the same circuits.

**SOFT Table 3 vs. exact values vs. Tsim GPU**
LER of kept shots (errors / kept); $A$ is the acceptance rate; Tsim brackets are 1000x likelihood intervals.
*Compute: 12 s GPU sampling + 0.3 s compile, 3.3e10 shots, 1 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)*

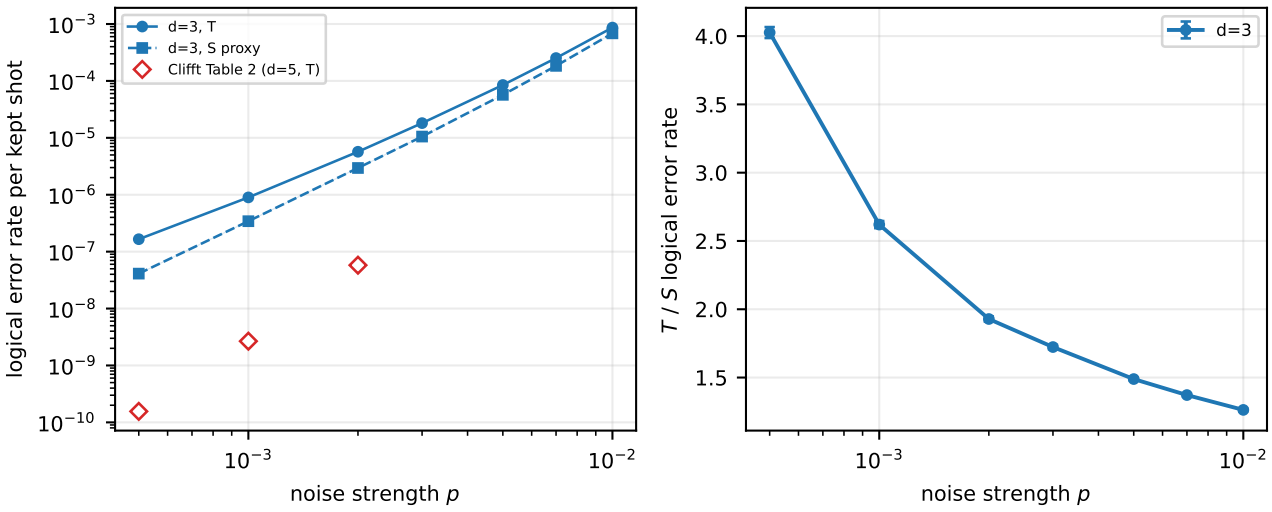
| $p$ | SOFT (Table 3) | exact (2609.18922) | Tsim GPU | shots, GPU time |
|---|---|---|---|---|
| 0.0005 | 1.57e-10 (8/5.1e10) | 2.846e-10, $A$=0.3792 | – | – |
| 0.001 | 4.62e-09 (49/1.1e10) | 3.328e-09, $A$=0.1440 | – | – |
| 0.002 | 3.67e-08 (22/6e8) | 5.175e-08, $A$=0.0208 | – | – |
| 0.003 | – | – | – | – |
| 0.004 | – | – | – | – |
| 0.005 | – | – | – | – |

# 4 Chase, Labib (2026), Clifft: fast exact simulation of near-Clifford quantum circuits

**arXiv:2604.27058** Study module `repro/studies/clifft_inject_cultivate.py`

Clifft's Fig. 6 plots the logical error rate per kept shot of Gidney's unitary-injection cultivation circuits (no escape) with real $T$ gates and with the $S$ proxy, and their ratio; its Table 2 lists the $d = 5$ $T$-gate values at three noise strengths (stratified importance sampling, $9.6 \times 10^{10}$ shots, 12 machine-hours on a 16-core c6i.8xlarge).

Clifft's circuits are the same as the Gidney study's (the Clifft paper's converter, byte-identical at $p = 10^{-3}$), so this study samples nothing itself: it re-plots the Gidney study's unitary-injection points as LER and $T/S$ ratio.




**Clifft Fig. 6: inject + cultivate, $T$ vs. $S$ proxy, and their ratio**

*Compute: (Gidney study points): 5.6 min GPU sampling + 4.0 s compile, 9e11 shots, 14 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)*

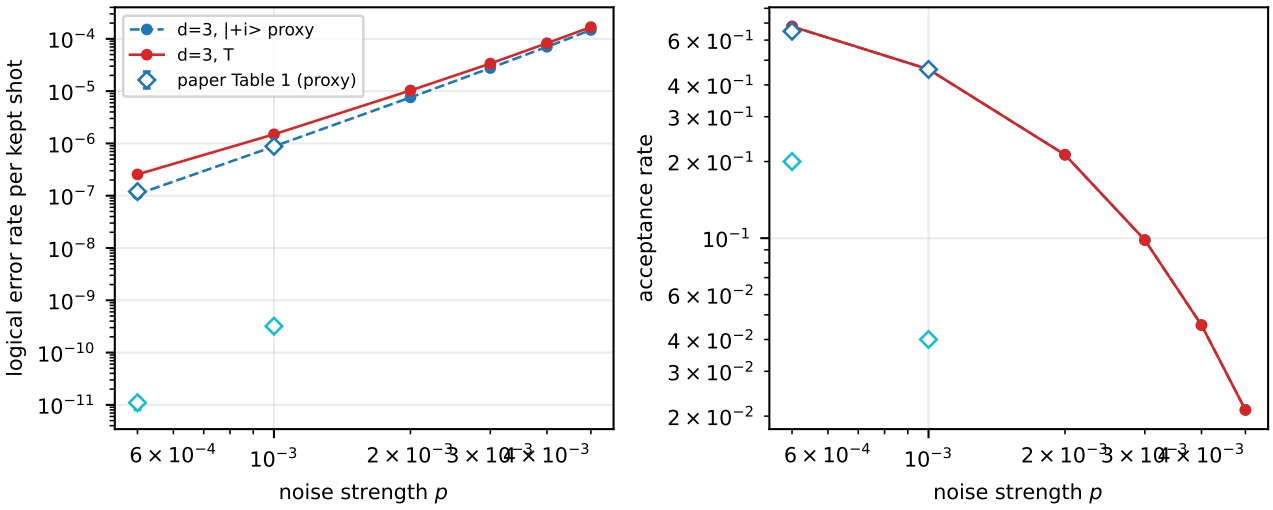
Same samples as the Gidney study (unitary injection). Error bars on the ratio are $\pm 1\sigma$ (Poisson). Open diamonds: Clifft's Table 2 $d = 5$ values (stratified importance sampling).

# 5 Hetényi, Brown, Williamson (2026), Constant-depth magic state cultivation with Clifford measurements by gauging

**arXiv:2603.05429** Study module `repro/studies/gauging_cultivation.py`

Fig. 4(a) compares, at $d = 3$ and $p = 0.1$–$0.5\%$, the authors' Stim simulation of the $|+i\rangle$ proxy with a qiskit state-vector simulation of the $T$ circuit; Table 1 lists $P_L$ and $p_{succ}$ for fault distances 3 and 5 at $p =$ 1e-3 and 5e-4 (the $d = 7$ entries are skipped here, as agreed). Every detector is postselected, including a final noiseless stabilizer round.

Circuits: the authors' generator `get_stim_circ_w_growing(d, basis="Y", pipeline=True)` with their `utils.add_noise` (uniform $p$), as in their notebook. Clifford proxy: as generated, final noiseless $Y_L$ readout. $T$ version: every $S/S^\dagger$ becomes $T/T^\dagger$ (as in the authors' qiskit check), and a noiseless logical rotation $\exp(\mp i\pi Z_L/8)$ (`repro/magic_check.py`) is inserted before the readout so that $Y_L$ checks the $T$ state deterministically, the same observable as the paper's per-shot $XS^\dagger$ expectation.




**Hetényi et al. Fig. 4(a,b) and Table 1: gauging cultivation, $d = 3$ and $d = 5$**
*Compute: 3.1 min GPU sampling + 4.2 s compile, 5.4e11 shots, 12 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)*
Full postselection including the final noiseless stabilizer round (counted in the success probability). Dashed: the $|+i\rangle$ Clifford proxy; solid: real $T$ gates with a noiseless $T$-basis readout. Open diamonds: the paper's Table 1 (proxy, Stim). The $d = 7$ point of the paper is not attempted.

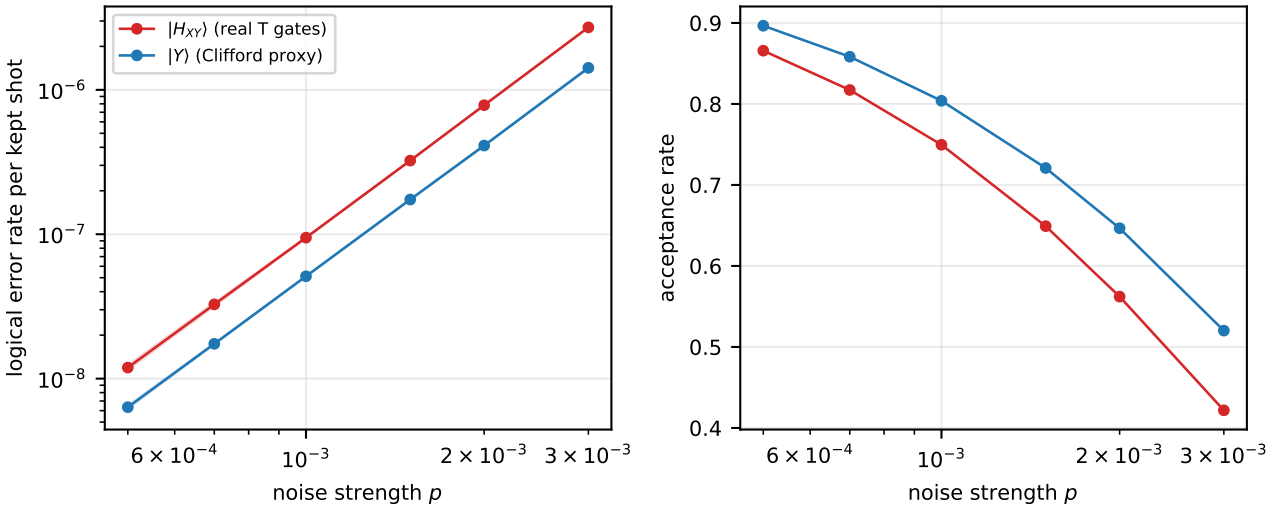
# 6 Sahay et al. (2025), Fold-transversal surface code cultivation

**arXiv:2509.05212** Study module `repro/studies/fold_transversal.py`

The paper compares $|Y\rangle$-state cultivation (Clifford, Stim) with the true $|H_{XY}\rangle = T|+\rangle$ cultivation (a state-vector simulation) on the fault-distance-3 surface-code protocol: unitary preparation plus stabilizer round, two fold-transversal $H_{XY}$ checks with GHZ(3) ancillas, then a noiseless decoding of the code and a logical readout; every detector postselected.

Circuits:

- Clifford proxy: the authors' `generate_sv_cirquit(p, dfinal=7, prep="unitstab", cirq=False)` (kaavyas99/MSC_ at 9378fd2), the circuit their state-vector handoff starts from; noiseless logical $Y$ readout.
- $T$: the native-$T$ reconstruction of the same circuit by Wan and Zapirain (arXiv:2609.18922, `fold-d3/model.json`: CY layers -> CCZ + controlled-$SX$), exported by our `make_circuits.py` at noise $p$, followed by a noiseless $T^\dagger$ and $X$ readout of the decoded logical qubit.




**Sahay et al. Fig. app_hxy(b): $|Y\rangle$ vs. $|H_{XY}\rangle$ cultivation, fault distance 3, no escape**

*Compute: 46.4 min GPU sampling + 9.7 s compile, 3.7e12 shots, 12 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)*

All detectors postselected; noiseless decoding and logical readout. The paper's figure is shown only as an image (no released numbers for this panel), so no overlay. As in the paper, the two states have different logical error rates once escape is excluded.

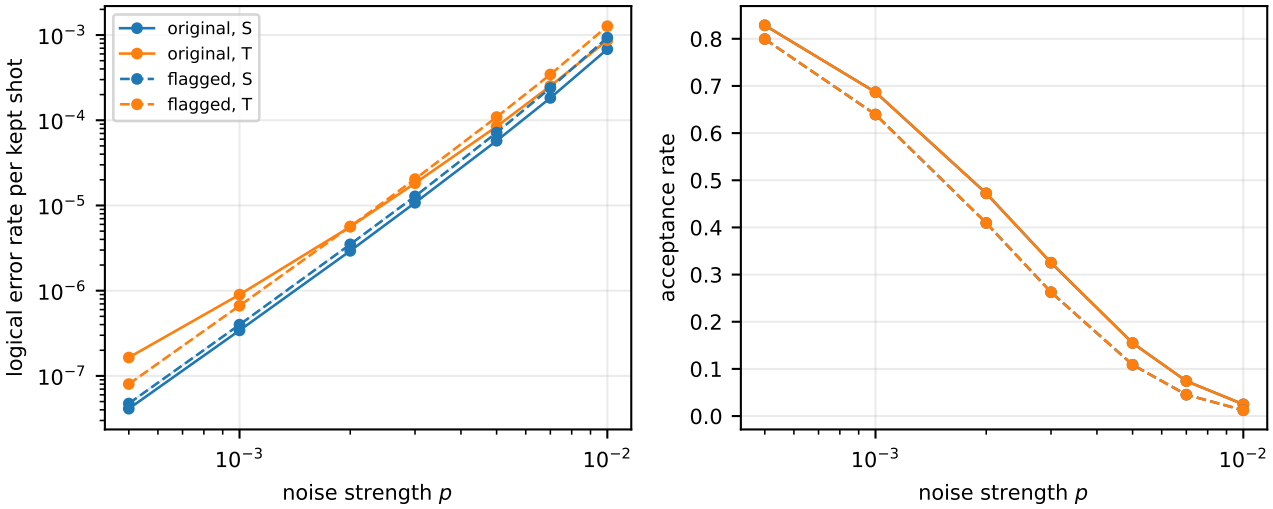
# 7 Chan, Strikis, Sun, Cai (2026), Diagnosing and restoring the degraded fault distance of magic state cultivation

**arXiv:2609.17706** Study module `repro/studies/chan_flags.py`

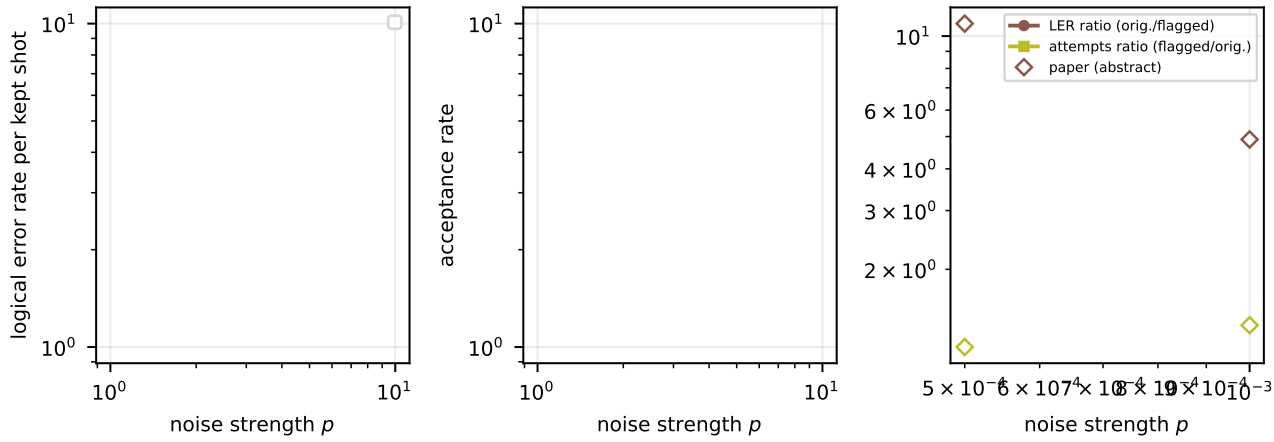
Pre-escape colour-code cultivation (injection + double checks, every detector postselected, noiseless final check), with and without the paper's Z flags in the final double check:

- Fig. `d3_pre_escape_ler`: $d = 3$ ($a = 6$ ancillas, $f = 2$ flags), $S$ vs. $T$;
- Figs. `d5_pre_escape_ler` and `improvement_and_cost_ratios`: $d = 5$ ($a = 19$, $f = 13$), $T$ state, original vs. flagged. The abstract quotes a 4.9x LER improvement at 1.36x attempts for $p = 10^{-3}$, and 10.9x at 1.17x for $p = 5 \cdot 10^{-4}$.

Circuits: the package's bundled $S$-state references (timchan0/cliffordea at 27385c6, `sim/stim_files`, the $d = 5$ ones with the Bell-growth feedforward and detector healing) and its own `make_variant_text`: `replace_noise_level` from $10^{-3}$ to $p$, and `swap_t_and_s_gates` for the $T$ state. The paper sampled with SymFT.




**Chan et al. Fig. d3_pre_escape_ler: $d = 3$ cultivation, $S$ vs. $T$, with and without Z flags**
*Compute: 19.9 min GPU sampling + 11 s compile, 1.2e12 shots, 28 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)*
Blue: $S$ proxy; orange: real $T$ gates. Solid: the original double check ($a = 6$); dashed: with two Z flags ($f = 2$). The flags restore fault distance 3 for the $T$ state.




**Chan et al. Figs. d5_pre_escape_ler and improvement_and_cost_ratios: $d = 5$ $T$ cultivation with Z flags**

*Compute: not yet sampled*

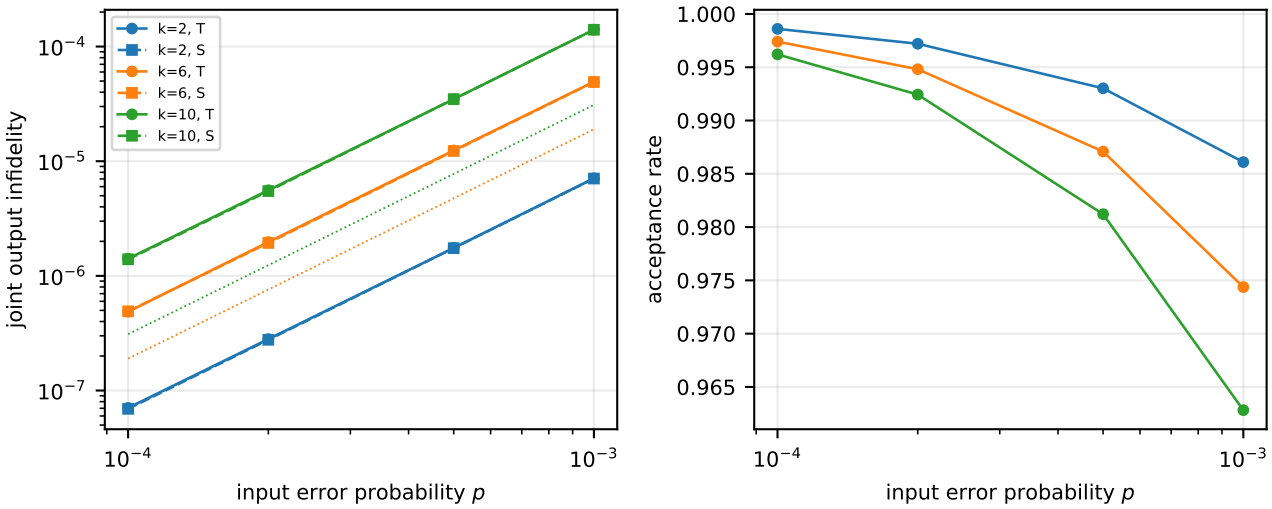
Pre-escape $d = 5$ $T$-state cultivation. Right: LER improvement and attempts ratio of the flagged circuit; open diamonds are the ratios quoted in the paper's abstract and Sec. VI.

# 8 Wang, Gu (2026), Magna: simulating magic state factories as cheaply as Clifford circuits

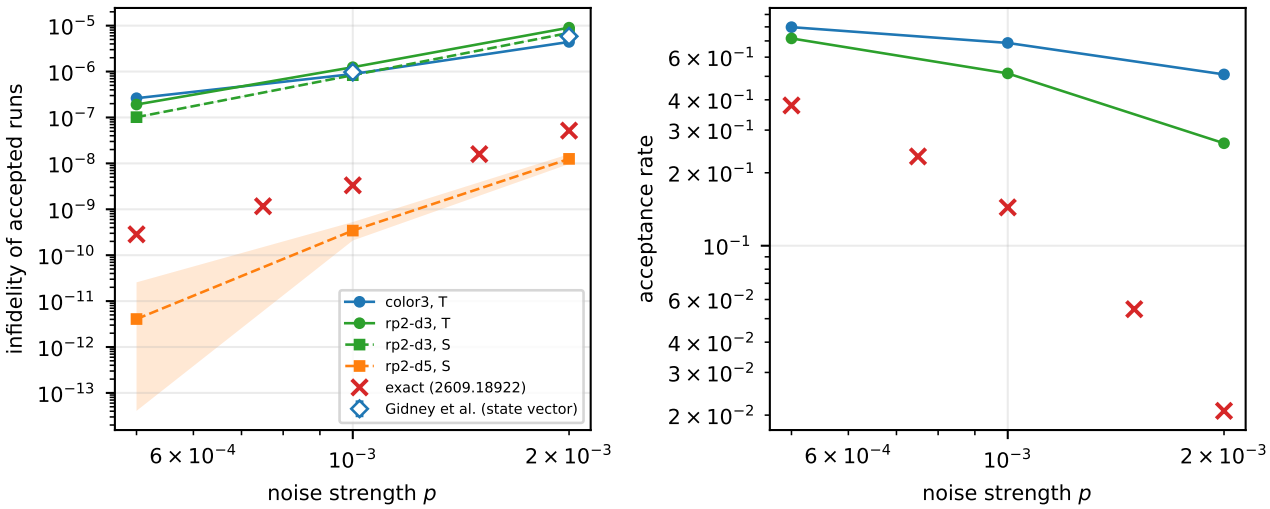
**arXiv:2610.22315** Study module `repro/studies/magna.py`

- Fig. 8(a): Bravyi–Haah $(3k+8)$-to-$k$ distillation on bare qubits, $k = 2, 6, 10$. Every input magic state has a $Z$ error with probability $p$; everything else is noiseless. Plotted: the joint infidelity of the $k$ outputs with $|T\rangle^{\otimes k}$, for $p$ from $10^{-4}$ to $10^{-3}$.
- Fig. 9(a,b): colour-code and $\mathrm{RP}^2$ cultivation at $d = 3, 5$ (no escape), acceptance rate and infidelity of accepted runs under uniform depolarizing noise.

Circuits: Magna's own example protocols (itpyi/magna at 01ae9c2, `examples/*/physical.magna`), translated to the Tsim dialect by our `make_circuits.py` (deferred-measurement controlled corrections etc.). The examples are written at $p = 10^{-3}$ and Magna ships no generator, so other noise strengths are made by changing that literal. Only the detectors named in the examples' `POSTSELECT` lines are postselected. The error check is the examples' task definition: undo the ideal target circuit on the output qubits and require every output to read 0 (a joint check). Clifford versions: Magna's `-S` examples (product-$S$ targets); for the colour code the $S$ proxy is in the Gidney study.




**Magna Fig. 8(a): Bravyi–Haah $(3k+8)$-to-$k$ distillation**
*Compute: 7.4 min GPU sampling + 8.2 s compile, 9e11 shots, 24 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)*
Bare qubits, $Z$ errors of probability $p$ on the input magic states only. Dotted: the leading order $(1+3k)p^2$. Dashed/squares: the matched Clifford circuit ($|S\rangle$ inputs, product-$S$ target).




**Magna Fig. 9(a,b): colour-code and RP$^2$ cultivation at $d = 3, 5$**
*Compute: 17.9 min GPU sampling + 4.9 s compile, 6.3e12 shots, 12 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)*
Magna's cultivation examples (no escape). Markers as in the paper: Gidney et al.'s state-vector values for Color $d = 3$ and the exact values of arXiv:2609.18922 for Color $d = 5$. Dashed: Magna's $S$-target RP$^2$ examples.

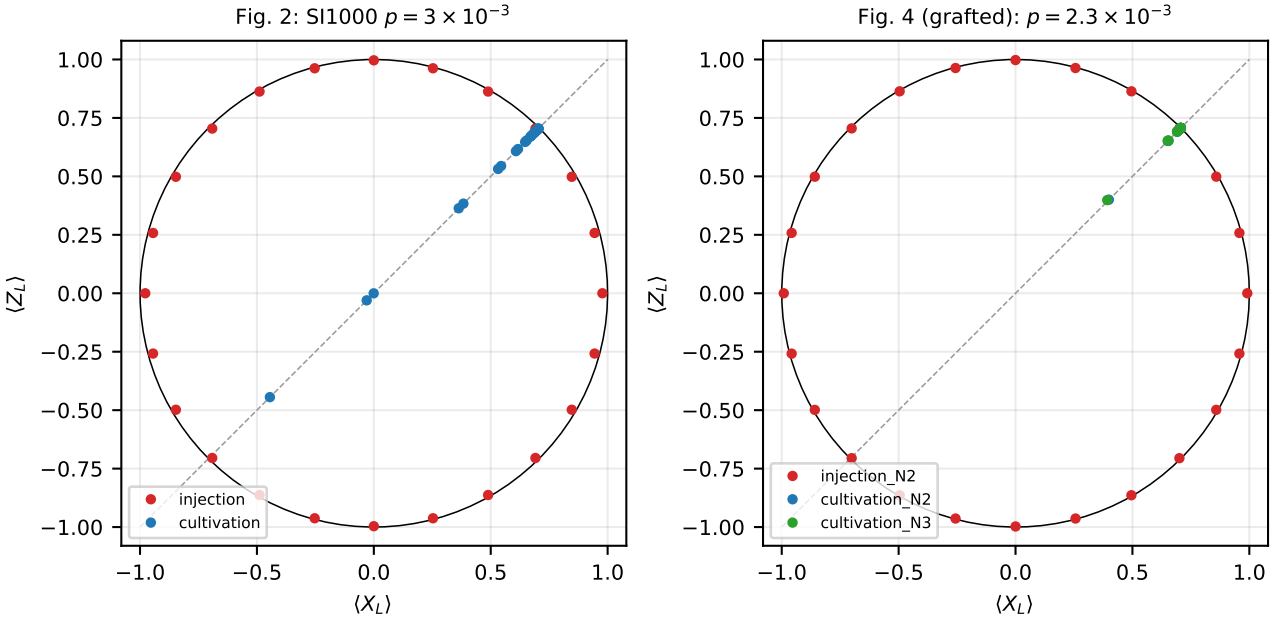
# 9 Rosenfeld et al. (2025), Magic state cultivation on a superconducting quantum processor

**arXiv:2512.13908** Study module `repro/studies/google_experiment.py`

The paper's grey points are state-vector simulations of the experimental circuits under the SI1000 noise model; shots are kept only if no detector fires (post-selected logical tomography). Here:

- Fig. 2(b,c): logical tomography ($\langle X_L\rangle$, $\langle Z_L\rangle$) of injection only and of injection + cultivation, vs. the injected angle $\theta$, SI1000 $p = 3.0 \times 10^{-3}$;
- Fig. 3(b): kickback tomography (KT) of the cultivated $|T\rangle$: logical error vs. $\theta$, with and without the QEC cycles, SI1000 $p = 2.3 \times 10^{-3}$; SI Fig. `validating_KT`: KT error at the $T$ angle vs. $p$ (expected $\propto p^3$);
- Fig. 4(b): tomography after grafting to the surface code, injection only and cultivation with $N = 2, 3$ cycles, SI1000 $p = 2.3 \times 10^{-3}$.

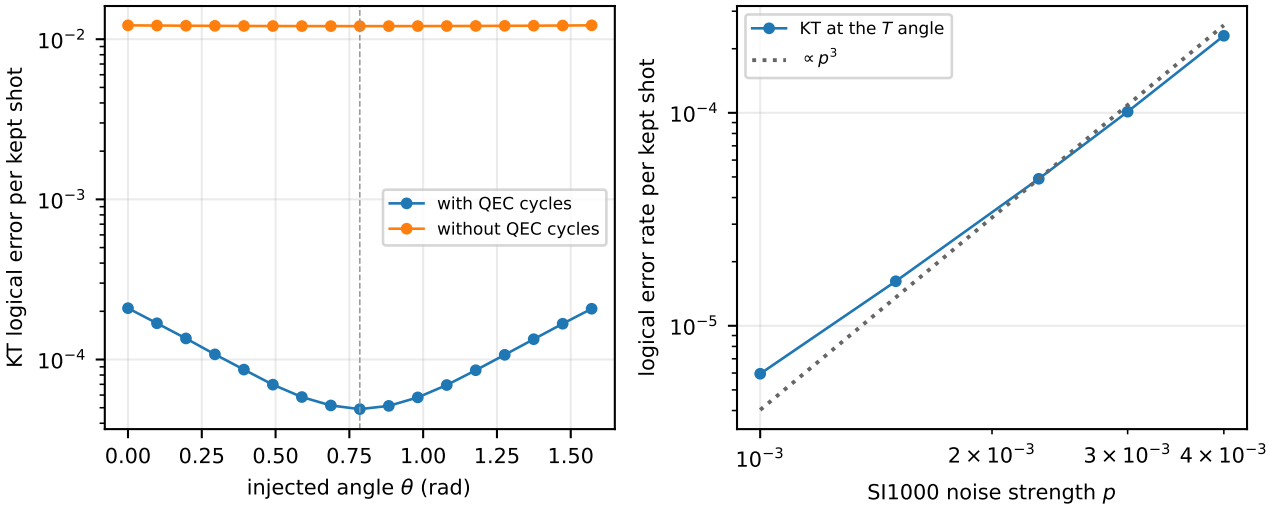
Circuits: the released Zenodo (17872003) circuits, repaired to real gates from their Cirq versions by our `make_circuits.py` (injection $e^{-i\theta Y/2}$ as an $H_{YZ}$-conjugated $Z$ rotation, $T/T^\dagger$ checks). The injection rotation is set to $R_Z(\theta/\pi)$; SI1000 noise is added with Gidney's `gen.NoiseModel.si1000(p)` (the model the paper cites), with non-Clifford gates mapped to Clifford placeholders while the noise is inserted. The experiment had no Clifford proxy, so there is none here.




**Rosenfeld et al. Figs. 2(b) and 4(b): post-selected logical tomography vs. injected angle**

*Compute: 2.1 min GPU sampling + 2.3 min compile, 7.2e11 shots, 240 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)*

Each marker is one injected angle $\theta$ (24 angles in $[0, 2\pi)$), with $\langle X_L\rangle$ and $\langle Z_L\rangle$ from the $X$- and $Z$-basis circuits (up to $3 \times 10^9$ shots each). Cultivation projects the state onto the magic axis (dashed). These replace the paper's grey state-vector points.



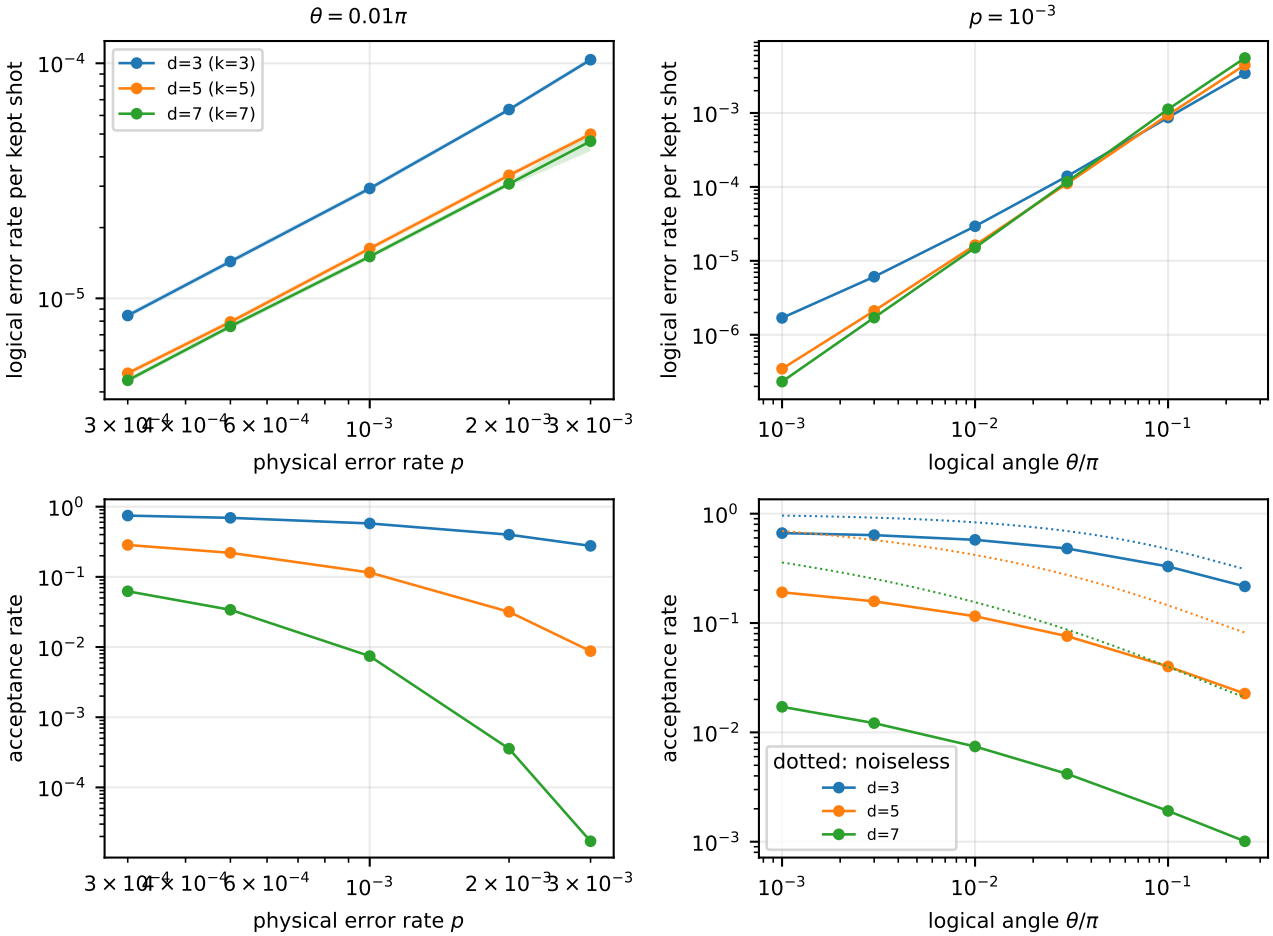

**Rosenfeld et al. Fig. 3(b) and SI Fig. validating_KT: kickback tomography of the cultivated $|T\rangle$**
*Compute: 2.3 min GPU sampling + 34 s compile, 7.6e11 shots, 38 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)*
Left: KT logical error vs. injected angle (SI1000 $p = 2.3 \times 10^{-3}$), with and without the interleaved QEC cycles. Right: KT error at $\theta = \pi/4$ vs. $p$, which scales as $p^3$ (fault distance 3).

# 10 Ismail et al. (2025), Transversal architecture for megaquop-scale quantum simulation with neutral atoms (transversal STAR)

**arXiv:2509.18294** Study module `repro/studies/star_injection.py`

The paper's Fig. 6 shows the logical error rate and success rate of its transversal multi-rotation (TMR) STAR injection with full postselection, vs. the physical error rate (at a logical angle of $10^{-2}$) and vs. the angle; Fig. 13 validates against an analytic success rate at $d = 3$.

The authors released no generator; the circuits here are QuEra's public STAR challenge circuits (`QuEra-Challenge/star_cir` with the challenge's own `physical_angle` helper to set the logical angle. They differ from the paper's Fig. 6 protocol: $k = d$ physical rotations (the paper uses $k = 3$; equal at $d = 3$), $d$ postselected rounds after the rotations, uniform depolarizing noise ($p = 10^{-3}$ in the released files; other $p$ by changing that literal), and a noiseless logical un-rotation and $X$ readout. Every detector is postselected (raw parity; the $d - 1$ detectors right after the rotations are random by design). The paper's Clifford approximation is not specified, so no Clifford version is sampled; the noiseless success probability $\cos^{2k}(\phi/2) + \sin^{2k}(\phi/2)$ of $k$ physical rotations by $\phi$ is overlaid.




**Ismail et al. Figs. 6, 13, 15 (protocol of the public STAR challenge circuits)**
*Compute: 20.4 min GPU sampling + 16 s compile, 7.6e12 shots, 30 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)*
Logical error of the injected small-angle rotation (noiseless un-rotation, logical $X$ readout) and success rate, with every detector postselected. Note the protocol differences listed above ($k = d$, $d$ postselected rounds after the rotations, uniform noise), so the curves are not expected to coincide with the paper's.

# 11 Chen et al. (2025), Efficient magic state cultivation on $RP^2$

**arXiv:2503.18657** Study module `repro/studies/rp2_ungrown.py`

Table 1 lists, at $p = 10^{-3}$, the logical error rate and discard rate of each cultivation process alone (no escape stage), benchmarked by sampling the Clifford variant ($S$ for $T$, cultivating $S|+\rangle$) with a final round of perfect stabilizer measurement and logical readout: MSC-3 $8.3 \times 10^{-7}$ / 48.6\%, MSC-5 $3.6 \times 10^{-10}$ / 90.5\%.

Circuits: the authors' `circuit_garage` files (Zihan-Chen-PhMA/Cultiv_T_RP2 at 77ad9b3, written at $p = 10^{-3}$ by their `d3_T_cult_sampling.py` / `d5_T_cult_sampling.py`). Clifford version: as released. $T$ version: $S \to T$, $S^\dagger \to T^\dagger$ (reversing the authors' substitution) plus a noiseless logical rotation (`repro/magic_check.py`) so that the final $Y_L$ readout checks the $T$ state deterministically. MSC-5 with $T$ gates has Tsim active width 10, which makes it the slowest circuit here.

**Chen et al. Table 1 (ungrown columns): $RP^2$ cultivation without escape,** $p = 10^{-3}$
Every detector postselected, perfect final stabilizer round and logical readout.
*Compute: 23 s GPU sampling + 0.5 s compile, 7.9e10 shots, 2 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)*

| | paper ($S$ proxy) | | Tsim GPU, $S$ proxy | | | Tsim GPU, $T$ | | |
|---|---|---|---|---|---|---|---|---|
| protocol | LER | discard | LER [1000x interval] (errors) | discard | shots, time | LER (errors) | discard | shots, time |
| MSC-3 | 8.3e-07 | 48.6% | 8.25e-07 [8e-07, 8.5e-07] (20095) | 48.60% | 4.7e10, 13 s | 1.25e-06 [1.2e-06, 1.3e-06] (20161) | 48.59% | 3.1e10, 9.6 s |
| MSC-5 | 3.6e-10 | 90.5% | – | – | – | – | – | – |

# 12 Jacoby et al. (2025), Magic state injection with erasure qubits

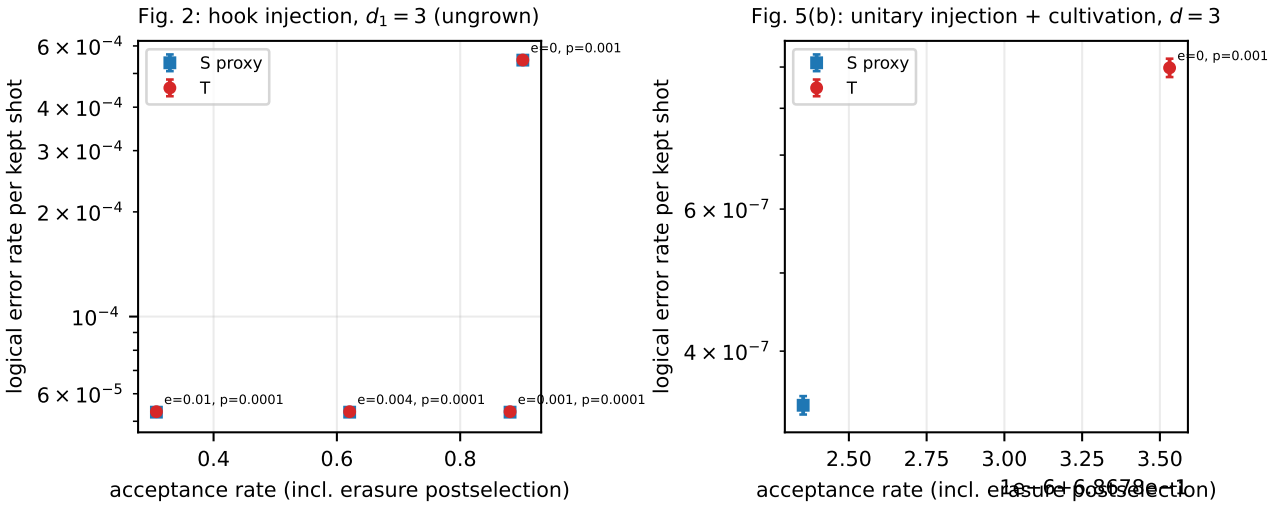
**arXiv:2504.02935** Study module `repro/studies/erasure_injection.py`

With erasure qubits, every heralded erasure in the (ungrown, fully postselected) patch discards the shot, so the logical error rate of kept shots depends only on the residual Pauli rate $p$, and the acceptance rate is the Pauli acceptance times the probability of no erasure. The paper's scenarios: $(e, p) = (0, 10^{-3})$, and $e \in \{10^{-3}, 4 \cdot 10^{-3}, 10^{-2}\}$ with $p = 10^{-4}$; erasure probability $e$ per two-qubit gate and SPAM operation and $e/10$ per single-qubit gate (`end2end_script.py`: `e1=e/10, e_SPAM=e`).

Circuits:

- Fig. 5(b), unitary injection + $d = 3$ colour-code cultivation: Gidney's generator with uniform depolarizing noise (the paper's circuit is Gidney's `inject[unitary]+cultivate` file), $S$ proxy and $T$; the $e = 0$ point is the Gidney study's $p = 10^{-3}$ point.
- Fig. 2, $d_1 = 3$ hook injection of $|S\rangle$: the released `circuits/QEC_hook_d=3.stim` ($p = 10^{-3}$; $p = 10^{-4}$ by scaling its noise literals, since the notebook that generated the paper's circuits is not released). $T$ version: the injection $S \to T$ and a noiseless logical rotation (`repro/magic_check.py`) so the final $Y_L$ readout checks the $T$ state. Lao–Criger injection and the hybrid layouts are not reproduced.

The no-erasure probability is computed exactly from the Pauli circuit's noise locations, following `get_erased_circuit`.




**Jacoby et al. Figs. 2 and 5(b): erasure qubits, ungrown injection and cultivation**

*Compute: 2.5 s GPU sampling + 1.3 s compile, 2.9e9 shots, 4 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)*

Each point is one erasure scenario $(e, p)$; the logical error rate comes from the Pauli-only circuit at residual rate $p$ (one sample serves every $e$), and the acceptance is the sampled Pauli acceptance times the exact no-erasure probability.

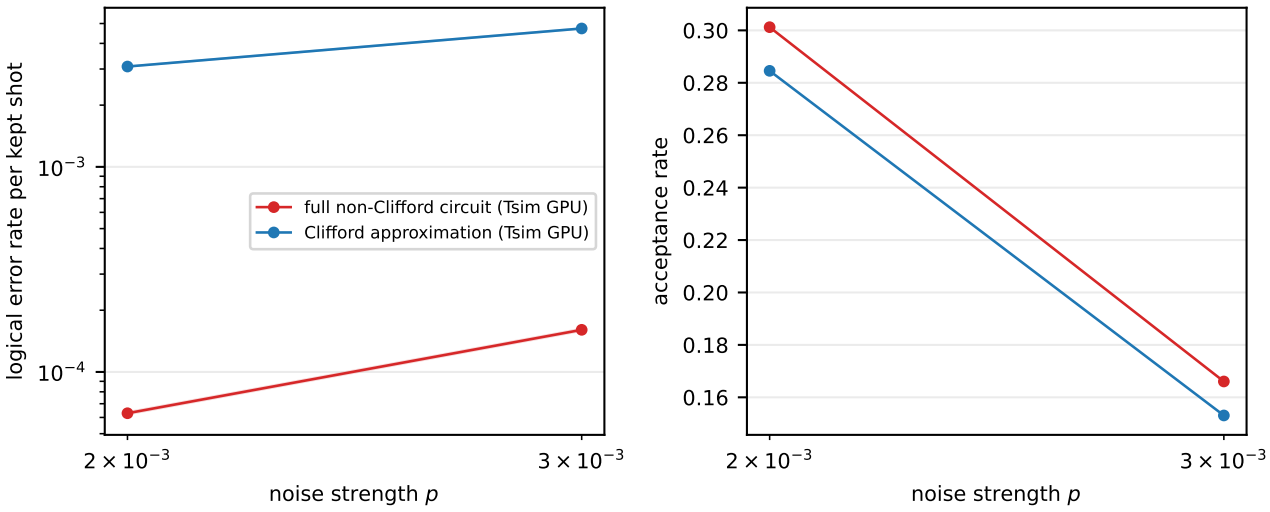
# 13 Vaknin et al. (2025), Efficient magic state cultivation on the surface code

**arXiv:2502.01743** Study module `repro/studies/vaknin_full_vector.py`

Unrotated $d_1 = 3$ surface-code $H$-type cultivation (injection + double-checking with GHZ ancillas), without the expansion step: every stabilizer measurement postselected, then a perfect logical measurement in the magic-state basis. The authors spent about $10^6$ CPU-hours on a full state-vector simulation, roughly $10^7$ shots per point, and compared it with their Clifford approximation.

Circuits: the authors' generator (tomirendo/surface-code-magic-state-cultivation at ec28f6b, `Simulation(d=3)`: `generate_syndrome_circuit(detect_x=False)` + `generate_injection_circuit` + `generate_double_ghz`) in its own pinned environment (`.venvs/vaknin`, `repro/external/vaknin_generate.py`), with their `NoiseModel().uniform_d`

- Clifford approximation: `to_stim_circuit(apply_non_cliffords=False)` (non-Clifford gates replaced, their noise kept), final $Z_L$ readout;
- non-Clifford: `apply_non_cliffords=True`, placeholders replaced by $T/T^\dagger$/CCZ/CH/CSWAP decompositions (our `make_circuits.py`); the final noiseless readout is set to $Z_L$, preceded by the noiseless logical rotation $\exp(\mp i\pi Y_L/8)$ (`repro/magic_check.py`) that turns the final $Z_L$ readout into the magic-basis ($H_L$) check.




**Vaknin et al. Fig. 26: unrotated $H$ cultivation, non-Clifford vs. Clifford approximation**
*Compute: 48 s GPU sampling + 11 s compile, 2.2e9 shots, 4 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)*
All stabilizers postselected, no expansion, perfect magic-basis logical readout. The paper's full state-vector run took about $10^6$ CPU-hours for $\sim 10^7$ shots per point; its numbers are shown only in the figure, so there is no overlay.